



SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences
Chennai- 602105



Student Name: B. HEMANTH KUMAR

Reg. No.:192324238

Course Code: CSA0911

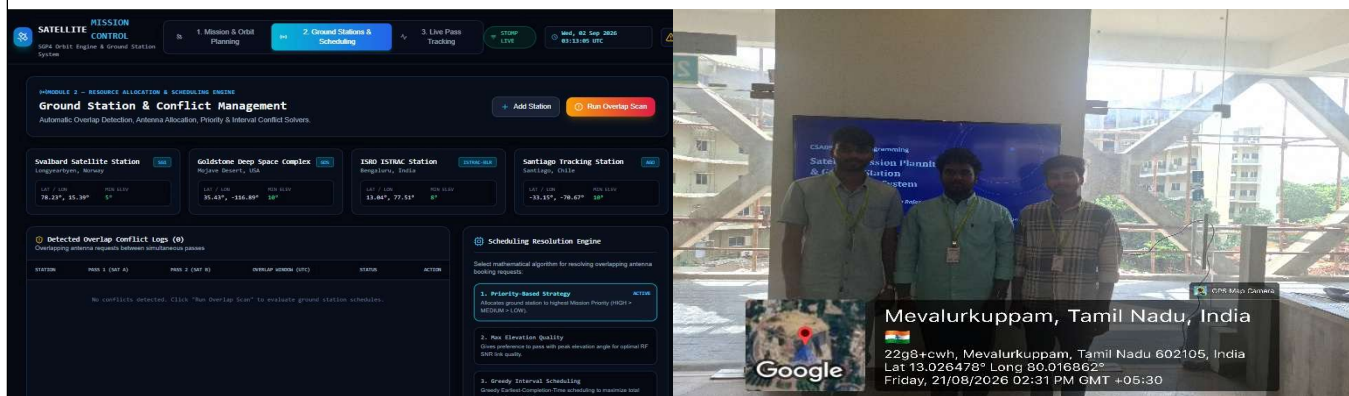
Slot: B

Course Name: JAVA PROGRAMMING

Course Faculty: Dr. R. THALAPATHI RAJASEKARAN

Project Title: SATELLITE MISSION PLANNING & GROUND STATION MANAGEMENT SYSTEM

Module Photographs: (3 photographs –Module Photo, Individual student contribution module work in the project and presentation image)



Project Description

The Satellite Mission Planning & Ground Station Management System is a software-based application designed to efficiently plan, schedule, monitor, and manage satellite missions and ground station operations. The system provides a centralized platform for managing satellite activities such as mission scheduling, satellite passes, ground station availability, communication windows, telemetry data, and command transmission.

The system allows mission operators to define satellite missions and generate optimized schedules based on factors such as satellite visibility, ground station availability, communication duration, mission priority, and operational constraints. It helps avoid scheduling conflicts and ensures that available ground station resources are utilized efficiently.

A Ground Station Management module is included to maintain information about ground stations, antenna availability, communication frequencies, operational status, and scheduled satellite contacts. The system can identify suitable ground stations for a satellite during its visible pass and provide operators with the required contact schedule.

Student Signature

Guide Signature